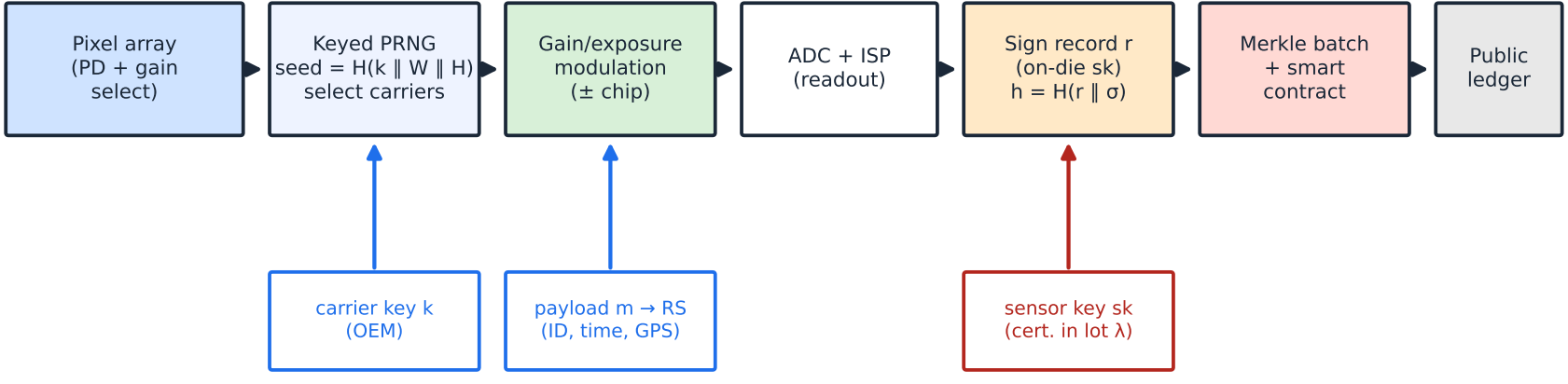
PixelRoot provenance-write datapath (capture-time)



Soft binding (in-pixel payload) + Hard binding (sensor-signed record, sealed on-chain) — bound together by the record's content fingerprint.